



Joe Power HDIP24
number 20109022
project ANDROIDS

Air Navigation Data for Radio Officer
Information Display System

The Operational Challenge

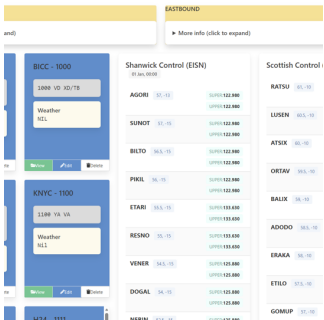


What is Shanwick Radio

Aeronautical Communications Centre for the North Atlantic.

HF and VHF Radio

Handles 1500 Aircraft Daily

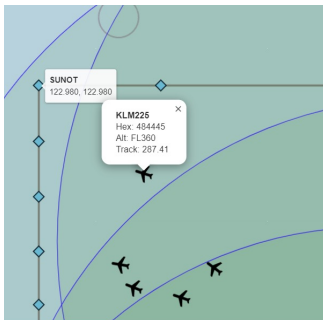


What is the IDS

HF Radio Station Frequencies and VHF ATC Centers

Weather, Nat Track,

Operation and Safety updates



What is ANDROIDS

Extends Map View .

Real Aircraft Data



Backend First

Separation of Concerns

A distinct split between backend and frontend (phase 1&2) ensures the system is scalable and extendable

Robust Backend

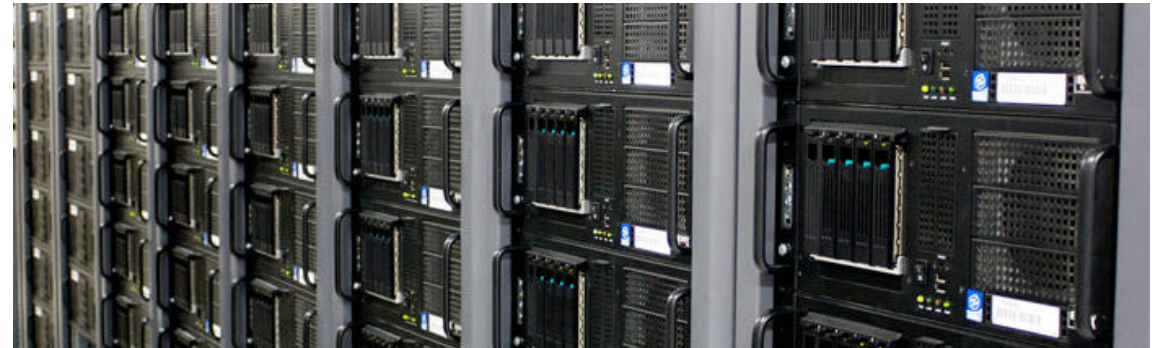
Hapi, Nodejs, Mongo DB

CRUD, Testing , Seeding

API and RBAC

Reliable and Secure

error handling, validation, and session authentication,



CYQX - 1000				BICC - 1000				EISN 01 Mar, 00:00				EGPX 01 Mar, 02:00			
1000	VB	XD		1000	VD	XD/TB		AGORI	37.33	SUPR	100.101	LOW	122.980		
1130	VB	VF	XD					SUNOT	37.33	SUPR	122.980	LOW	122.980		
Weather NIL				Weather BIRD 02 VOLMET				BILTO	34.5.15	SUPR	122.980	LOW	122.980		
View Edit Delete				View Edit Delete				PIKIL	34.5.15	SUPR	122.980	LOW	122.980		
								ETARI	34.5.15	SUPR	133.630	LOW	133.630		
								RESNO	34.5.15	SUPR	133.630	LOW	133.630		
								VENER	34.5.15	SUPR	125.880	LOW	125.880		
								DOCAI	34.5.15	SUPR	125.880	LOW	125.880		
								RATSU	34.5.15	HIGH	111.999	LOW	124.70		
								LUSEN	34.5.15	HIGH	133.630	LOW	124.70		
								ATSIX	34.5.15	HIGH	133.630	LOW	124.70		
								ORTAV	34.5.15	HIGH	133.630	LOW	124.70		
								BALIX	34.5.15	HIGH	132.730	LOW	124.70		
								ADODO	34.5.15	HIGH	133.630	LOW	124.70		
								ERAKA	34.5.15	HIGH	133.630	LOW	124.70		
								ETUC	34.5.15	HIGH	133.630	LOW	124.70		



Modelling Concepts

User Stories and Roles

Roles like Admin, Editor, and Viewer to shape requirements.

System

Backend Manual Input, Joi

Frontend fetch JSON, API.

Workflows and Interfaces

Sketched out pages and navigation.

Understand Screen Layout and environment

Assessing Data Models

Needs to pass testing and schema

Refined Center , Simplified Nat Tracks or deleted Levels

Finding the Data Model

Flight Information Regions Structure

FIRs (Centers) has multiple sectors .

Sector

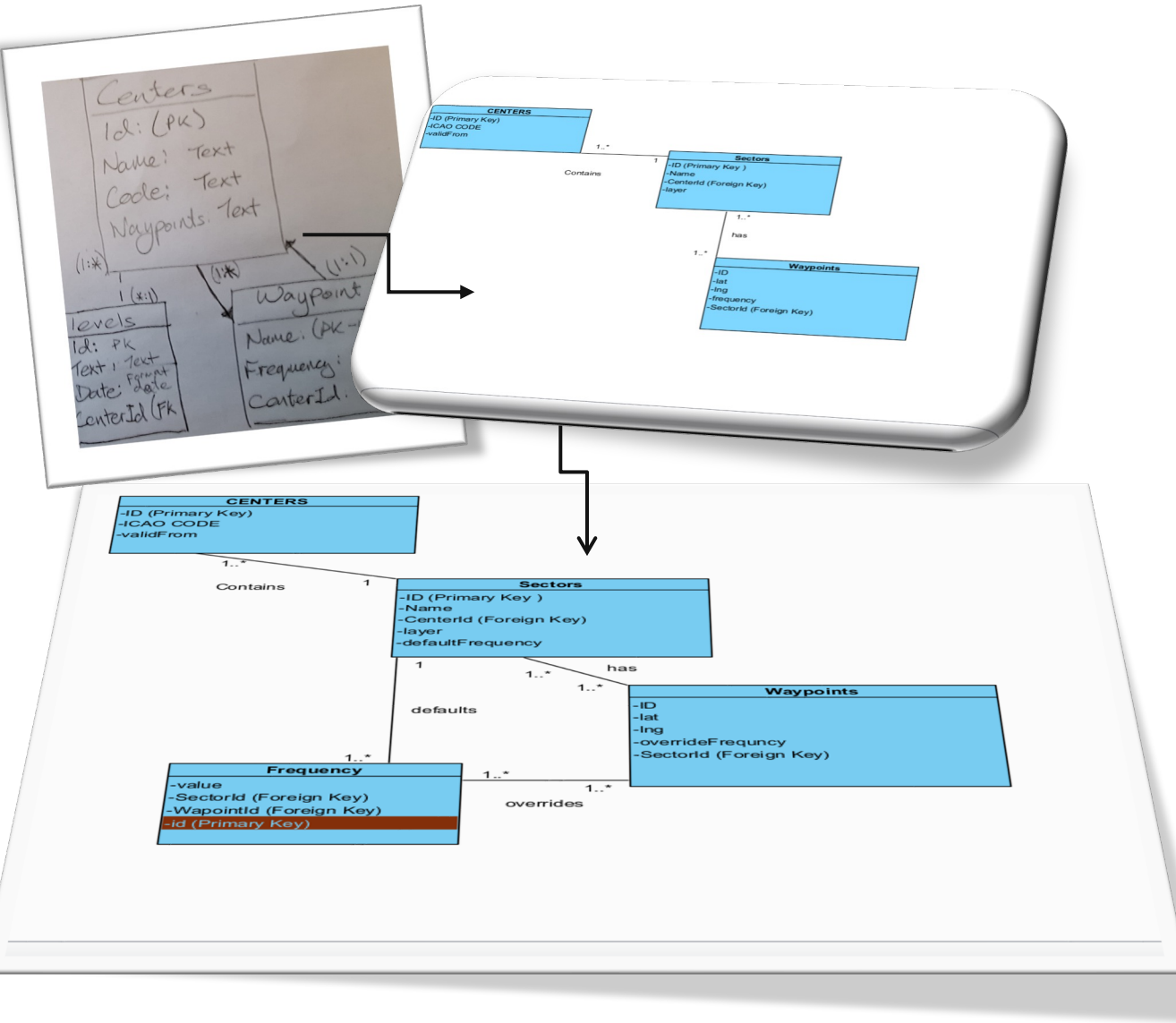
Sector sub space of Center. Divided by area and flight level.

Waypoints

Waypoints for navigation within a sector

Normalisation and Scalability

The model is normalised for scalability and prevents duplication. Frequency relates to Sector and Waypoint

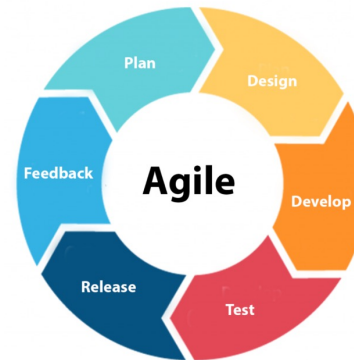
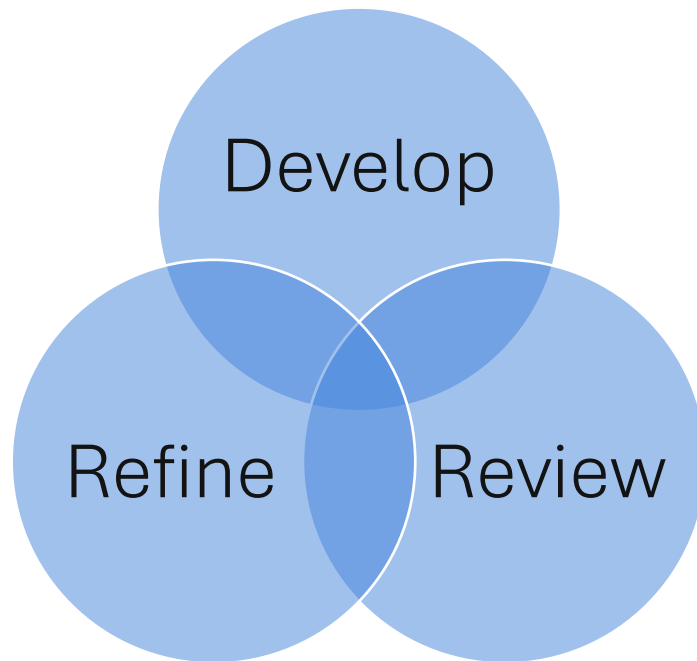


Implementation Overview



Minimum Viable Product Focus

Coursework as guide to basic requirement
Role Based access control
Helpers



Iterative Development Process

Supervisor and colleague Feedback
Agile planning once MVP attained
Sigmet instead of Space Weather
Simplify if needed (NAT TRACKS)



Fortify Process

GitHub branches. Merge. Tag. Stash.
Consistency. MVC
Confirm with Model and API
Database integrity

Project Roadmap

Research

similar systems. Hardware and network

Planning

Tech Stack Selection. Modelling

Early Sprints

Core Fundamentals CRUD AUTH

Supervisor / Interim

Conclude Phase 1, Report and log

Remodelling

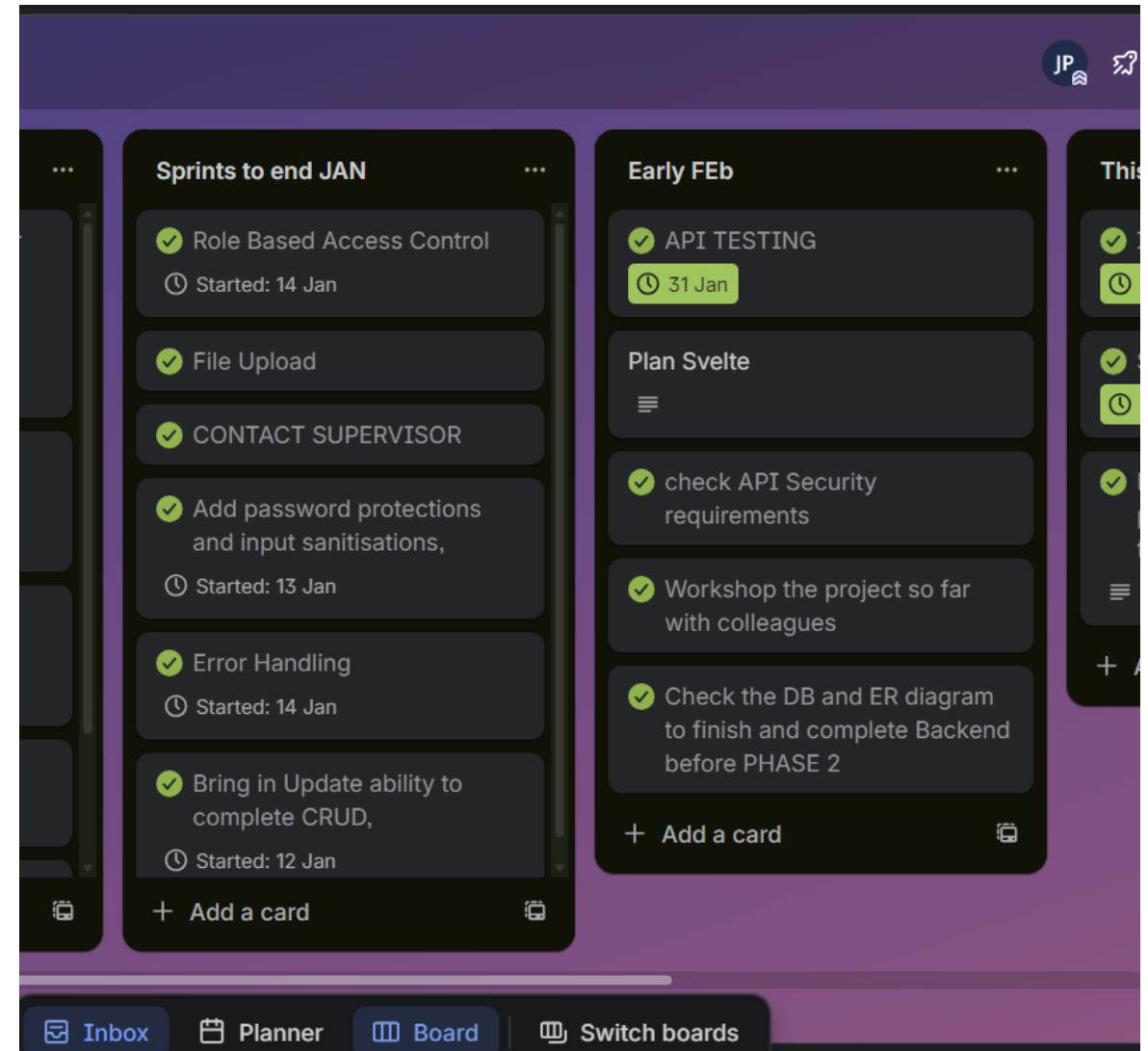
Improving Model, continue Backend

Phase 2

Less meticulous planning . Base Map then add

Timeline Drift

Work Based Project



Svelte (PHASE2)

Backend Data

Geospatial data has been integrated.

API-driven data supports both maps and lists.

Frontend

Layer Control, Legend, Boundaries, Space Weather, and Aircraft tracking.

VHF distance calculation.

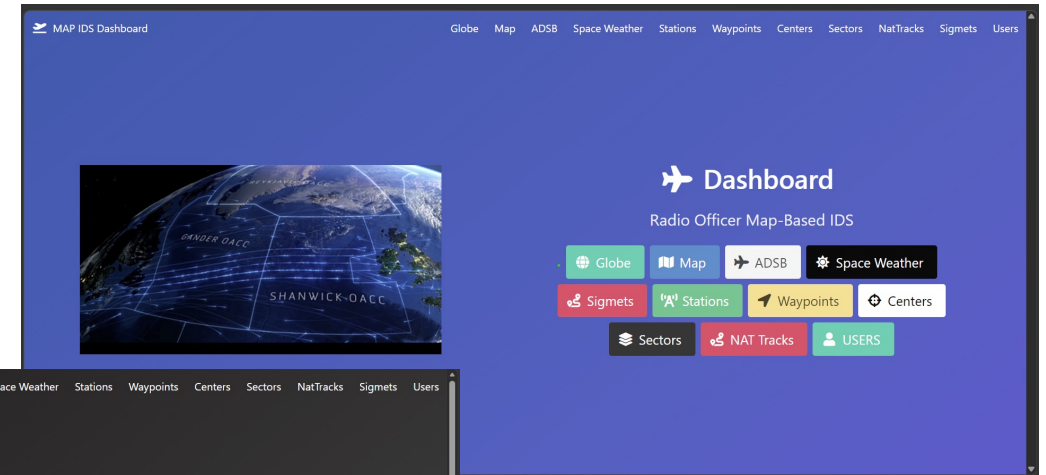
TODO

TypeScript

Dynamic data requiring continuous updates (e.g., ADSB, Space Weather).

Components and services

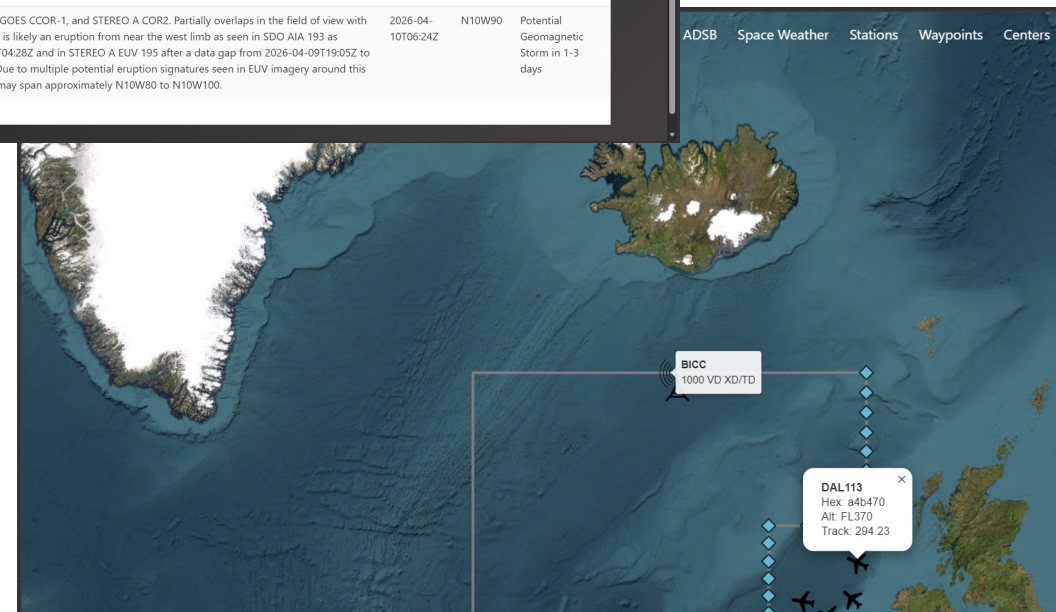
A more effective display panel .



Space Weather

Type	Strength	Time	Location	HF Impact
Solar Flare	M1.0	2026-04-09T08:45Z	N01W76	Possible HF Balckout on sunlit side
Coronal Mass Ejection	CME observed to the W in STEREO A COR2, GOES CCOR-1 after an eclipse period ends, and later frames of SOHO LASCO C2/C3 due to a data gap between 2026-04-10T04:22Z and 2026-04-10T08:06Z. The source is near AR 14408 (N08W83), observed in SDO AIA 131/171/193/311/304 as field line movement and brightening as early as 2026-04-10T03:42Z with further movement of field lines at 2026-04-10T05:09Z and 2026-04-10T06:27Z, followed by post-eruptive brightening over AR 14408. STEREO A EUVI returns from a data gap around 2026-04-10T05:45Z missing most of the main eruption, but observes post-eruptive field line movement and brightening in STEREO A EUVI 195/304 from this area near AR 14408, well north of AR 14409.	2026-04-10T06:09Z	N08W83	Potential Geomagnetic Storm in 1-3 days
Coronal Mass Ejection	CME visible to the SW in SOHO LASCO C2/C3, GOES CCOR-1, and STEREO A COR2. Partially overlaps in the field of view with CME: 2026-04-10T06:09Z. The potential source is likely an eruption from near the west limb as seen in SDO AIA 193 as moving field lines starting around 2026-04-10T04:28Z and in STEREO A EUV 195 after a data gap from 2026-04-09T19:05Z to 2026-04-10T05:55Z as post-eruptive arcades. Due to multiple potential eruption signatures seen in EUV imagery around this time, the exact source location is unclear, and may span approximately N10W80 to N10W100.	2026-04-10T06:24Z	N10W90	Potential Geomagnetic Storm in 1-3 days

No geomagnetic storm data

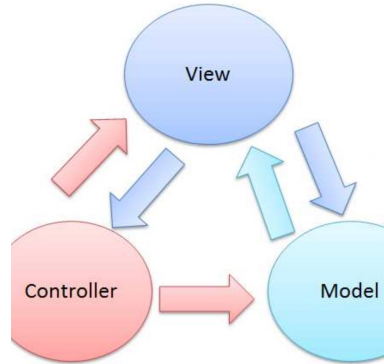


Project Development Insights

Co Pilot /IDE

Innovation
/Perseverance

Coursework/
References



Base Code

Vast majority derived from coursework or referenced
Example Full Stack I accessed over 65 times
Online references, docs over 30

Creative Structures

Niche project, Innovative necessary
Default and reset, not just CRUD
Leaflet instead of Cesium (2D over 3D)
Perseverance in commit history

AI assistance

Unhelpful for architecture
console to render referenced in Appendices.
Must be controlled and verified

Conclusion

Current

MVP almost achieved.

Frontend still in dev

Work in progress.

Nat Tracks (Model) and CRUD VHF frequencies.

Typescript and Login

Future

Oceanic ADSB coverage, Search, Alerts

Hosted

Air Gapped, Cloud, possible VPN

Challenges

Regulations, API Security, Site Acceptance, Maintenance

